

Arduino and sensor BPM HW-487 with simply amplifier

Before 2 years I've started my tests with this BPM sensor HW-487 according tutorial by these (aktualised) links:

https://navody.drtek.cz/navody-k-produktum/arduino-senzor-tepu-srdce.html?_gl=1*10acvhf*_gcl_au*R0NMLjE3NzkyODU2NzEuQ2p3S0NBand0N1hRQmhCa0Vpd0F0U3RwcDNNaTFybXNocGJTdV9RRncyDhHdVBpankxSDRQR1BEWmpfQmNWaEhiQ2ExaTR3aFlCd3pSb0NSd3NRQXZEX0J3RQ..*_gcl_au*MTQwODYyNTg5Ni4xNzc3NjY1MzA5
<https://drtek.cz/arduino-platforma/918-senzor-tepu-srdce.html>

But my succses with results were mostly bad. So I've started tests with another types of BPM sensores . In tat number I've tested the HW-827 reflexion BPM sensor and I have found, that in this sensor is the DC level eliminated due the schema I've founded on webs.

So I've tried to use the similar way for the improving of the useful signal from HW-487 BPM sensor. I've tried to use the simpliest amplifier with one NPN transistor and the DC signal from HW-487 was separed with capacitor 1 M and the signal from this transistor was connect direct with A0 analog input of Arduino Nano (in my case – note that HW-487 need 5V power for well DC output signal from it !!).

There is principial schema of my BPM sensor with HW-487, amplifier with NPN transistor and Arduino Nano (for example) :

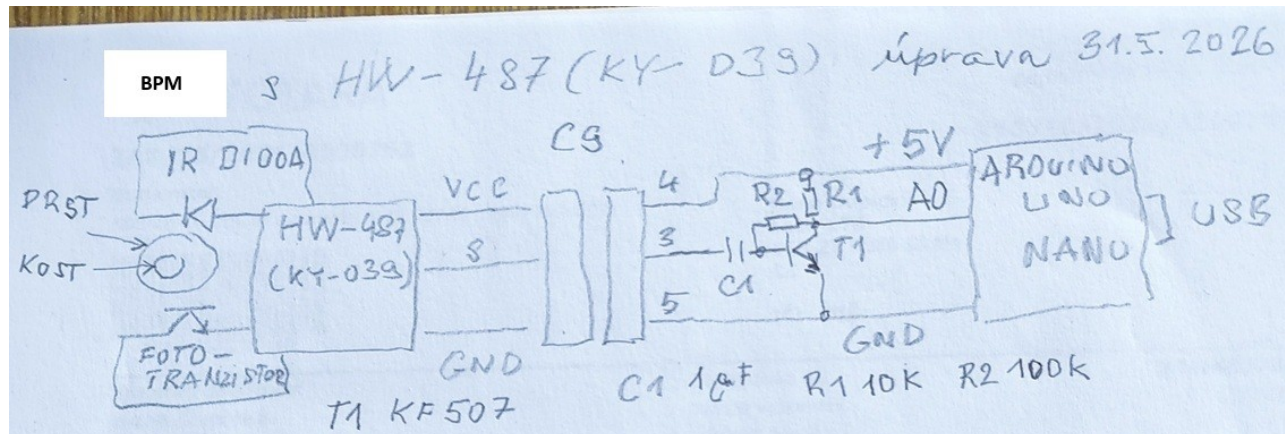


Fig. 1 - BPM sensor with HW-487, amplifier with transistor and Arduino Nano (for example)

There are on the next page photos of my wood BPM sensor with HW-487 before and after next my improoving of one :

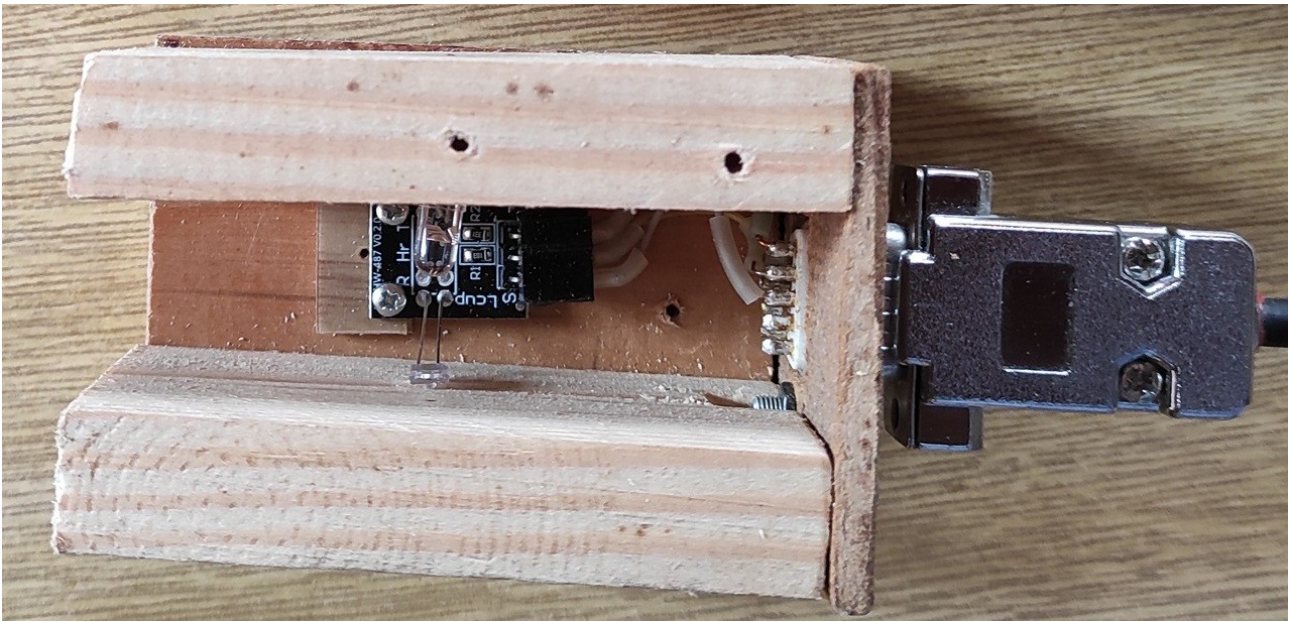


Fig. 2 - Wood BPM sensor with HW-487 before next improving

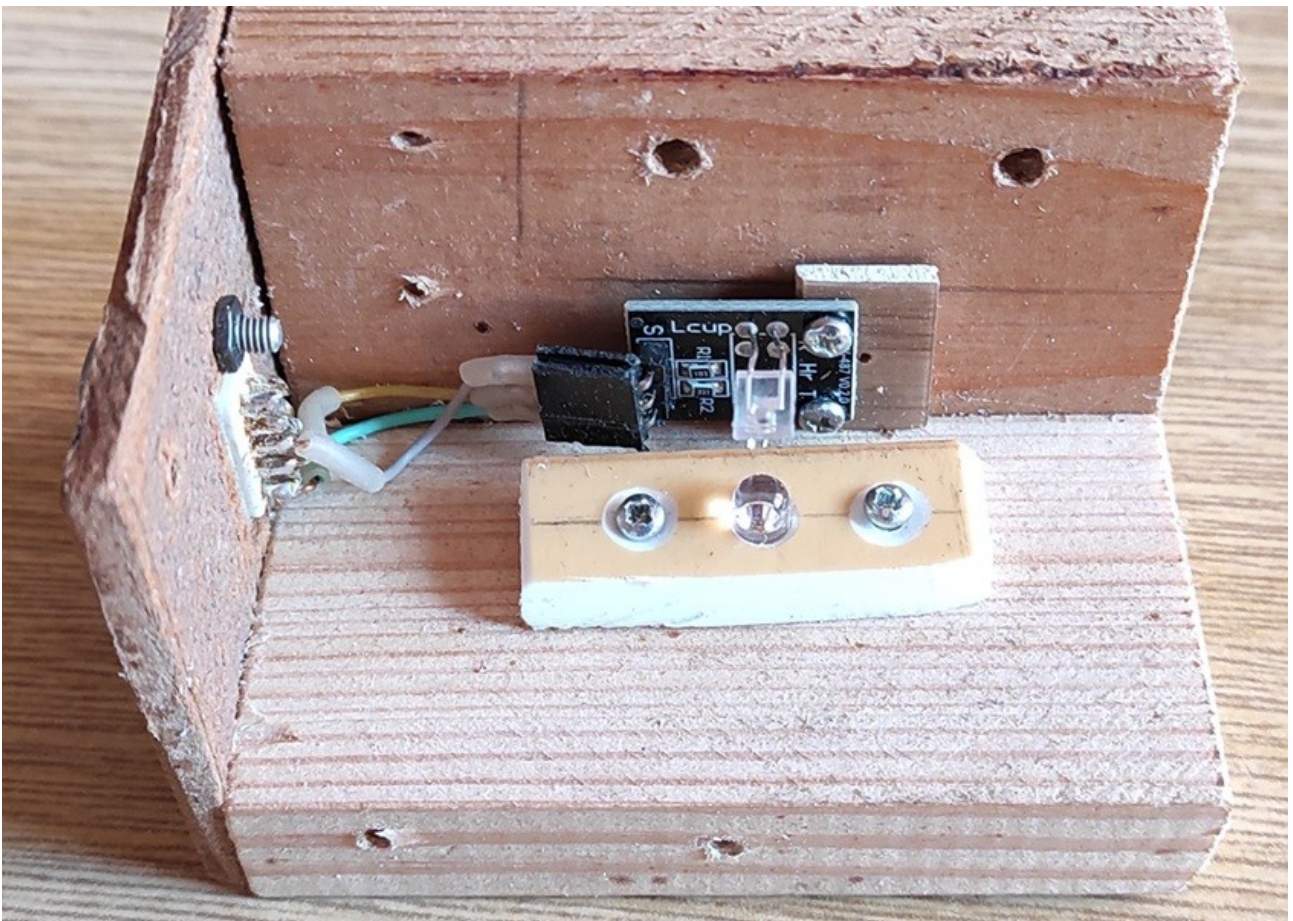


Fig. 3 - Wood BPM sensor with HW-487 with next important improving

There are on the next page photos of my BPM measurement system with HW-487, amplifier with transistor KF597 and Arduino Nano, all coupled with cable with Canon C9 connectors :

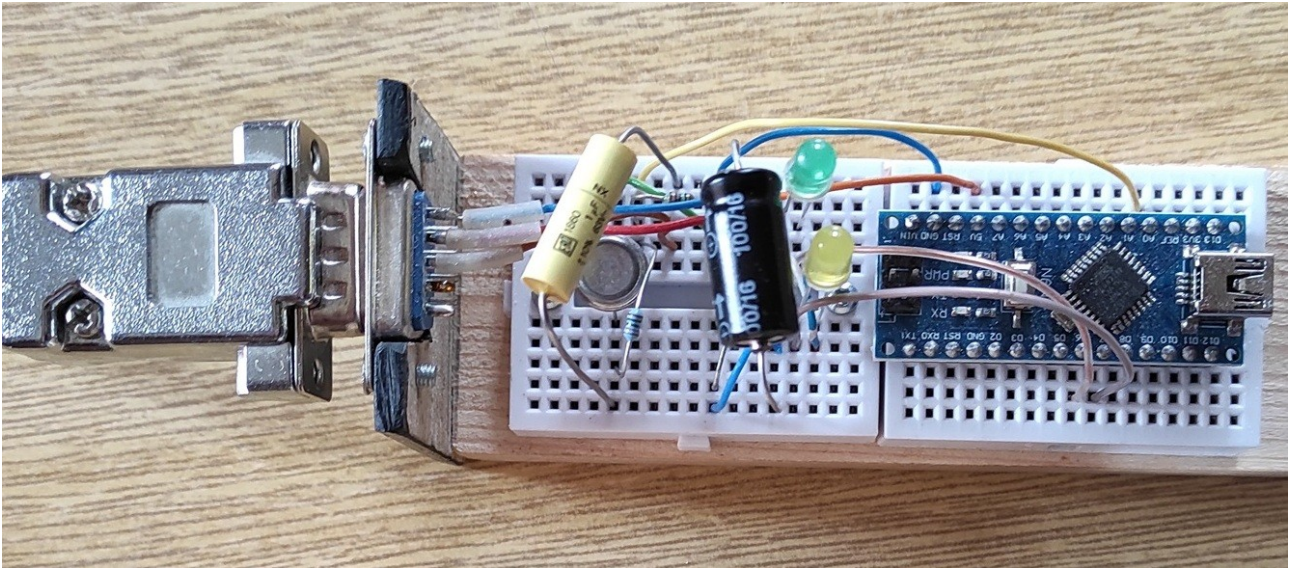


Fig. 4 – The realization of amplifier with NPN transistor and with Arduino Nano
(there used LEDs are available example)

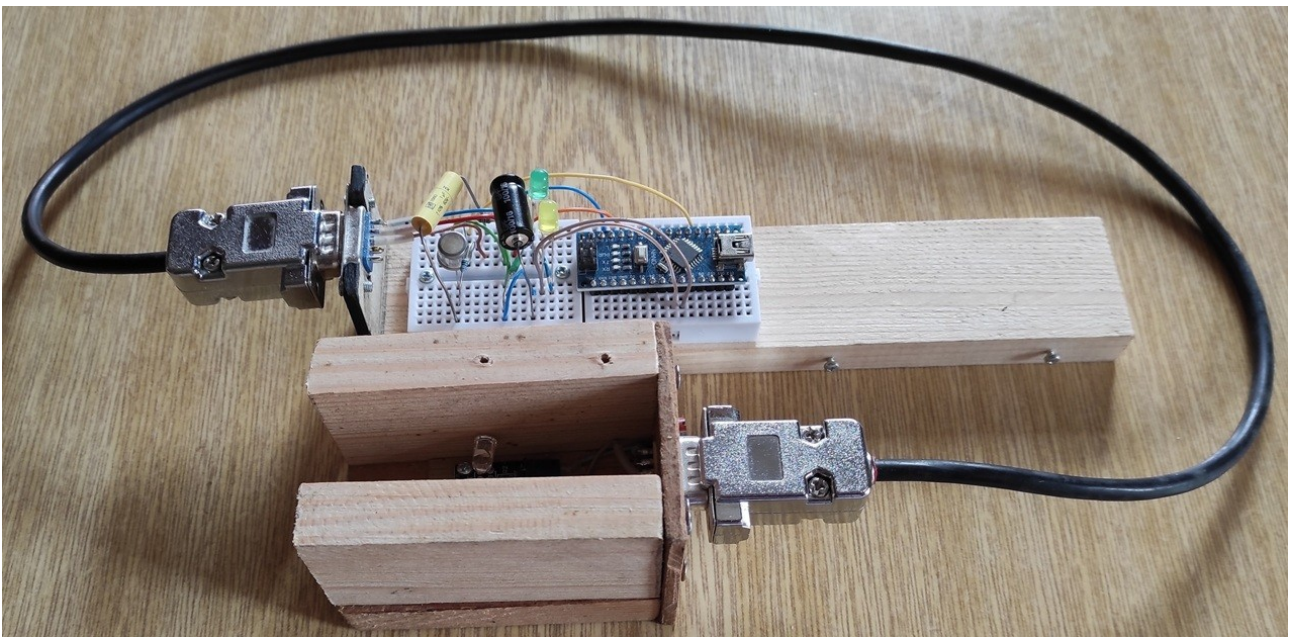


Fig. 5 – The connection of BPM HW-487 sensor with Arduino Nano

There is possible to use for example any sketch for reading serial data from Arduino via USB to a graphical plotter for example the one in Arduino IDE or more sophisticated program system.

For this purpose I've programmed for Arduino sketch **BPM_senzor_HW_487_en_3.ino** and for OS Windows **SerialPlotterEcg10.exe** both ones free for Github.

My program is named ...Ecg10.exe, as there is possible to read with it a serial data from another Ecg sensors such as AD8232 with Esp8266 for example (3.3 V !!) .

There are on the next page two examples of graphical and tabularised results from my exe program:

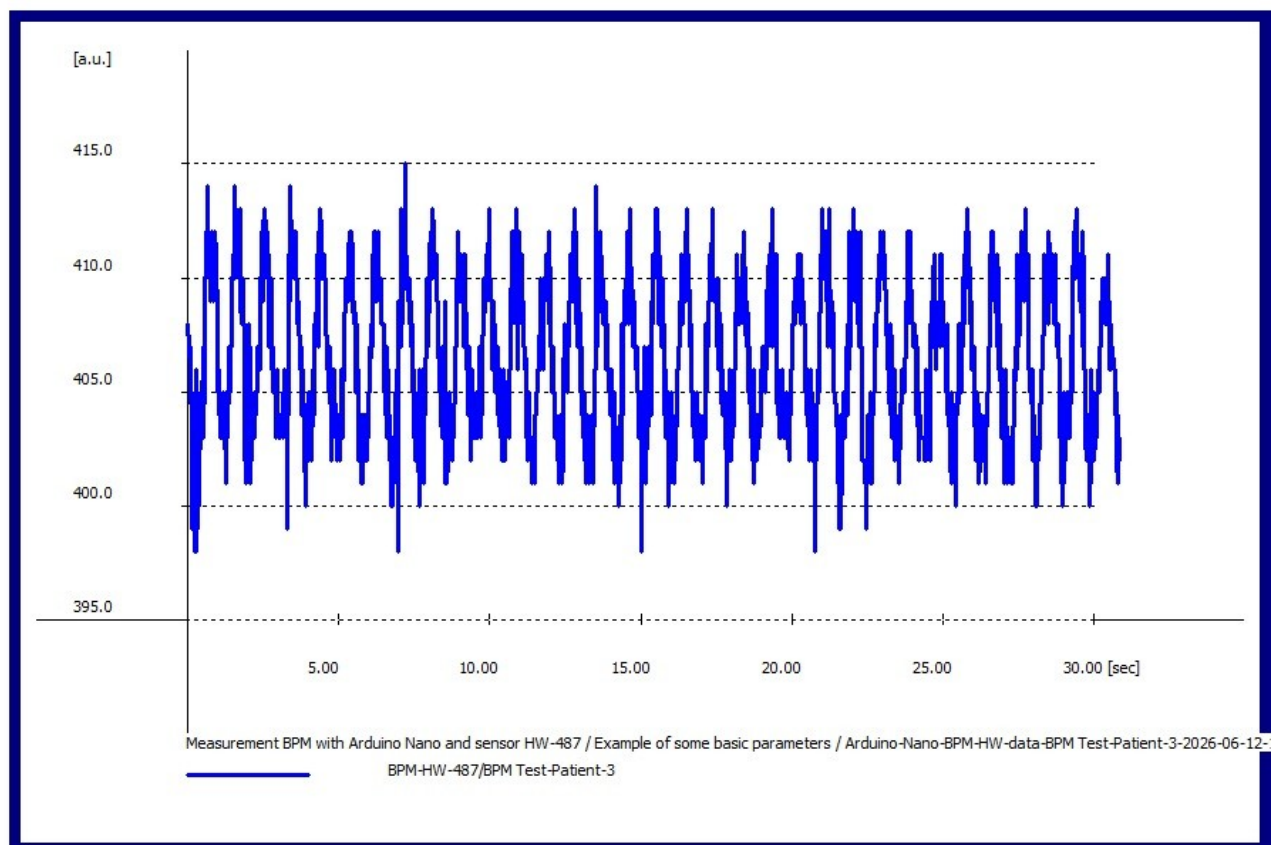


Fig. 6 – The graphical result from measuring BPM with HW-487 sensor and Arduino Nano

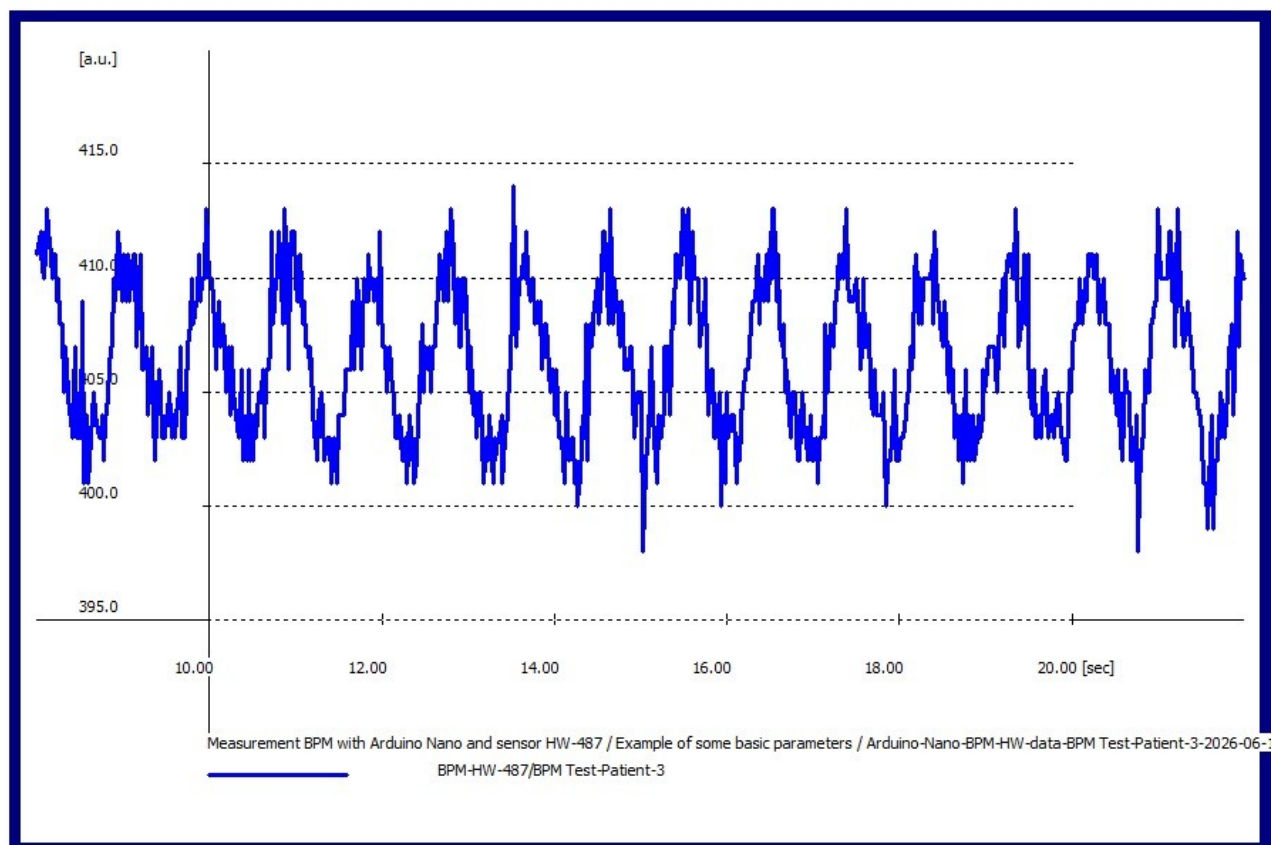


Fig. 7 – Detatil of graphical result from BPM measuring

Tabelized results from previous measurement published in previous graphs:

Arduino-Nano-BPM-HW-data-BPM Test-Patient-3-2026-06-12-123006.tab

File : Arduino-Nano-BPM-HW-data-BPM Test-Patient-3-2026-06-12-123006.txt

Bpm-Arduino-Nano+HW-Com3-9600-example.par

Measurement BPM with Arduino Nano and sensor HW-487

Example of some basic parameters

Data: from 100, to 1550, all 1628,

Parameters: Data min 398, Data max 415, Threshold 410, Beat time limit 0.4sec,

Bpm: Nr of Bpm beats: 30, Bpm time count: 27.45/0.000, BPM: 66 /min,

Bpm health result: OK+(1)

Inputs Selected (list) :

Inputs 3 : 3/BPM-HW-487,BPM Test-Patient-3,a.u.,0,Merani Bpm

Bpm health time diferencies:

Index|Btime|Delta |Result

1	0.929	+0.032	OK
2	0.950	+0.053	OK
3	1.023	+0.126	OK
4	0.832	-0.065	OK
5	0.903	+0.006	OK
6	0.947	+0.050	OK
7	0.934	+0.037	OK
8	0.946	+0.049	OK
9	0.838	-0.059	OK
10	1.123	+0.226	OK
11	0.814	-0.083	OK
12	0.864	-0.033	OK
13	1.028	+0.131	OK
14	0.854	-0.043	OK
15	0.946	+0.049	OK
16	0.948	+0.051	OK
17	0.883	-0.014	OK
18	1.075	+0.178	OK
19	0.926	+0.029	OK
20	0.802	-0.095	OK
21	0.929	+0.032	OK
22	0.961	+0.064	OK
23	0.948	+0.051	OK
24	0.899	+0.002	OK
25	0.965	+0.068	OK
26	0.883	-0.014	OK
27	0.899	+0.002	OK
28	0.882	-0.015	OK
29	0.975	+0.078	OK

end

Note: Results from measuring BPM with sensor HW-487 and Arduino Nano via program SerialPlotterEcg10.exe are only for personal use and they are not a medical documents.

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